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MEDICAL SENSOR, AND SENSOR SUPPORTING MEMBER

Abstract

At least one light emitter is configured to emit light. At least one light detector is configured to output a signal corresponding to an amount of incident light. A supporting member is adapted to be attached to a body of a subject. The supporting member includes a first supporting portion, a second supporting portion and a third supporting portion. The first supporting portion supports and covers the light emitter. The second supporting portion supports and covers the light detector. The third supporting portion is located between the first supporting portion and the second supporting portion. At least one of the first supporting portion, the second supporting portion and the third supporting portion includes a portion with higher flexibility than an adjacent portion thereof.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application is based on Japanese Patent Application No. 2024-037120 filed on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure relates to a medical sensor provided with a light emitter that emits light, and a light detector that outputs a signal corresponding to an amount of incident light. The present disclosure also relates to a sensor supporting member for attaching the medical sensor to a body of a subject.

[0003] Japanese Patent Publication No. 2020-150996A discloses, as an example of a medical sensor, a sensor adapted to be attached to a fingertip of a subject to calculate a transcutaneous arterial oxygen saturation (SpO₂) of the subject. The sensor includes a light emitter and a light detector. The light emitter emits light toward a tissue of the fingertip. The light having passed through the tissue is received by the light detector. The light detector outputs a signal corresponding to intensity of the detected light. The sensor includes a supporting member adapted to be wrapped around the fingertip. The supporting member supports the light emitter and the light detector. The signal that is outputted from the light detector is transmitted through a cable to a device that calculates SpO₂.

SUMMARY

[0004] It is demanded to improve the convenience of a medical sensor configured such that a light emitter and a light detector are attached to a subject's body using a supporting member.

[0005] A first illustrative aspect of the present disclosure may provide a medical sensor, comprising: [0006] at least one light emitter configured to emit light; [0007] at least one light detector configured to output a signal corresponding to an amount of incident light; and [0008] a supporting member adapted to be attached to a body of a subject, [0009] wherein the supporting member includes: [0010] a first supporting portion supporting and covering the light emitter; [0011] a second supporting portion supporting and covering the light detector; and [0012] a third supporting portion located between the first supporting portion and the second supporting portion; and [0013] wherein at least one of the first supporting portion, the second supporting portion and the third supporting portion includes a portion with higher flexibility than an adjacent portion thereof.

[0014] A second illustrative aspect of the present disclosure may provide a sensor supporting member adapted to attach, to a body of a subject, a medical sensor provided with at least one light emitter configured to emit light and at least one light detector configured to output a signal corresponding to an amount of incident light, the sensor supporting member comprising: [0015] a first supporting portion supporting and covering the light emitter; [0016] a second supporting portion supporting and covering the light detector; and [0017] a third supporting portion located between the first supporting portion and the second supporting portion, [0018] wherein at least one of the first supporting portion, the second supporting portion and the third supporting portion includes a portion with higher flexibility than an adjacent portion thereof.

[0019] The first supporting portion is a portion that supports and covers the light emitter. According to the configuration of each of the first and second illustrative aspects, elongation of the first supporting portion by covering the light emitter is promoted, and generation of a stress in the supporting member, and misalignment of the light emitter due to the stress can be suppressed.

[0020] The second supporting portion is a portion that supports the light detector while covering the light detector. According to the configuration of each of the first and second illustrative aspects, elongation of the second supporting portion due to the coverage of the light detector is promoted,

so that it is possible to suppress generation of a stress in the supporting member as well as the misalignment of the light emitter due to such stress.

[0021] The third supporting portion is a portion that is to be bent in order to cause the light emitter and the light detector to face each other. According to the configuration of each of the first and second illustrative aspects, not only the supporting member can be bent with a smaller force, but also the maintainability of the bent state is enhanced. As a result, it is possible to suppress misalignment of the light emitter and the light detector after the medical sensor is fixed to the body of the subject.

[0022] Therefore, it is possible to improve the convenience of the medical sensor configured such that the light emitter and the light detector are attached to the body of the subject with the supporting member.

[0023] A third illustrative aspect of the present disclosure may provide a medical sensor, comprising: [0024] at least one light emitter configured to emit light; [0025] at least one light detector configured to output a signal corresponding to an amount of incident light; and [0026] a supporting member adapted to be attached to a body of a subject, [0027] wherein the supporting member includes: [0028] a first retaining portion having a first adhesive face adapted to be adhered to the body; and [0029] a second retaining portion having a second adhesive face adapted to be adhered to a surface of the first retaining portion while covering the surface, [0030] wherein at least a part of the surface includes a portion with lower adhesiveness than an adjacent portion thereof.

[0031] A fourth illustrative aspect of the present disclosure may provide a sensor supporting member adapted to attach, to a body of a subject, a medical sensor provided with at least one light configured to emit light and at least one light detector configured to output a signal corresponding to an amount of incident light, the sensor supporting member comprising: [0032] a first retaining portion having a first adhesive face adapted to be adhered to the body; and [0033] a second retaining portion having a second adhesive face adapted to be adhered to a surface of the first retaining portion while covering the surface, [0034] wherein at least a part of the surface includes a portion with lower adhesiveness than an adjacent portion thereof.

[0035] When the medical sensor is fixed to the body of the subject, the second adhesive face is adhered to the surface of the first retaining portion. Since a portion having low adhesiveness is formed on the surface, the peelability of the second retaining portion from the first retaining portion can be enhanced. As a result, it is possible to facilitate the operation of re-attaching the sensor in order to, for example, optimize the arrangement of the light emitter and the light detector. Therefore, it is possible to improve the convenience of the medical sensor configured such that the light emitter and the light detector are attached to the body of the subject with the supporting member.

Description

DESCRIPTION OF THE DRAWINGS

[0036] FIG. 1 illustrates an appearance of a front face side of a sensor according to an exemplary embodiment.

[0037] FIG. 2 illustrates an appearance of a back face side of the sensor of FIG. 1.

[0038] FIG. 3 illustrates how to attach the sensor of FIG. 1 to a finger of a subject.

[0039] FIG. 4 illustrates how to attach the sensor of FIG. 1 to the finger of the subject.

[0040] FIG. 5 illustrates how to attach the sensor of FIG. 1 to the finger of the subject.

[0041] FIG. 6 illustrates an exemplary cross-sectional structure of a supporting member in a region VI illustrated in FIG. 1.

[0042] FIG. 7 illustrates another exemplary cross-sectional structure of the supporting member in the region VI illustrated in FIG. 1.

[0043] FIG. **8** illustrates an appearance of a front face side of a sensor according to another exemplary configuration.

[0044] FIG. **9** illustrates an exemplary cross-sectional structure of a supporting member in a region IX illustrated in FIG. **8**.

[0045] FIG. **10** illustrates another exemplary cross-sectional structure of the supporting member in the region IX illustrated in FIG. **8**.

[0046] FIG. **11** illustrates an appearance of a sensor according to another exemplary configuration.

DESCRIPTION OF EMBODIMENTS

[0047] Examples of embodiments will be described below in detail with reference to the accompanying drawings. In each of the drawings, the scale is appropriately changed in order to make each element as illustrated have a recognizable size.

[0048] FIG. **1** illustrates an appearance of a sensor **10** according to an exemplary embodiment. The sensor **10** is attached to a fingertip of a subject to calculate a blood light absorber concentration. The sensor **10** is an example of a medical sensor. The fingertip is an example of a body of a subject. Examples of the blood light absorber concentration include a percutaneous arterial oxygen saturation (SpO₂).

[0049] The sensor **10** includes at least one light emitter **11**. The light emitter **11** is a semiconductor light emitter that emits light including a prescribed wavelength. Examples of the semiconductor light emitter include a light emitting diode and a laser diode. The number of light emitter **11** and the value of the wavelength are appropriately determined in accordance with the type of the blood light absorber concentration to be acquired.

[0050] The sensor **10** includes at least one light detector **12**. The light detector **12** outputs a detection signal corresponding to an amount of light detected on a light detecting face. Examples of the light detector **12** include a photodiode, a phototransistor, and a photoresistor.

[0051] The sensor **10** includes a cable **13**. The sensor **10** is connected to a pulse photometer (not illustrated) through the cable **13**. The cable **13** accommodates a power supply line, a control signal line, and a detection signal line.

[0052] The power supply line and the control signal line are connected to a driving circuit (not illustrated) that is connected to the light emitter **11**. The power supply line supplies power from the pulse photometer to the driving circuit. The control signal line supplies a control signal transmitted from the pulse photometer to the driving circuit for controlling the activation or deactivation of the light emitter **11**.

[0053] The power supply line is also connected to the light detector **12**. Accordingly, the electric power from the pulse photometer is also supplied to the light detector **12** through the power supply line. The detection signal line electrically connects the light detector **12** and the pulse photometer. Accordingly, the detection signal outputted from the light detector **12** is transmitted to the pulse photometer, and subjected to appropriate processing.

[0054] The sensor **10** includes a supporting member **14**. FIG. **1** illustrates a front face side of the supporting member **14**. FIG. **2** illustrates a back face side of the supporting member **14**.

[0055] As illustrated in FIG. **1**, the supporting member **14** has a first supporting portion **141**, a second supporting portion **142**, and a third supporting portion **143**. The first supporting portion **141** supports and covers the light emitter **11**. The second supporting portion **142** supports and covers the light detector **12**. The third supporting portion **143** is located between the first supporting portion **141** and the second supporting portion **142**.

[0056] In the initial state, the back face side of the supporting member **14** is covered with a peelable sheet (not illustrated). When a user peels the sheet upon the usage of the sensor **10**, an adhesive face **144** illustrated in FIG. **2** is exposed. The adhesive face **144** has adhesiveness.

[0057] The adhesive face **144** includes a first translucent portion **144a** and a second translucent portion **144b**. The first translucent portion **144a** is disposed so as to face the light emitter **11**. The second translucent portion **144b** is disposed so as to face the light detector **12**.

[0058] Each of the first translucent portion **144a** and the second translucent portion **144b** is formed of a material having transparency with respect to the wavelength of the light emitted from the light emitter **11**.

[0059] A method of attaching the sensor **10** to a finger **20** of a subject will be described with reference to FIGS. **1** to **5**.

[0060] As illustrated in FIG. **3**, the first supporting portion **141** is disposed so as to face a nail side of the finger **20** under the condition that the adhesive face **144** of the supporting member **14** is exposed by the peeling of the sheet on the back face side. The adhesive face **144** located on a back face side of the first supporting portion **141** is adhered to the nail side of the finger **20**.

[0061] As illustrated in FIG. **1**, the supporting member **14** includes a first retaining portion **145**. The first retaining portion **145** extends leftward and rightward of the first supporting portion **141**. As illustrated in FIG. **2**, the adhesive face **144** includes a first adhesive face **144c**. The first adhesive face **144c** is located on a back face side of the first retaining portion **145**.

[0062] As illustrated in FIG. **3**, the first retaining portion **145** is wrapped around the finger. As a result, the first adhesive face **144c** is adhered to the finger **20**.

[0063] Subsequently, the supporting member **14** is bent while the third supporting portion **143** is faced to a tip of the finger **20**, whereby the second supporting portion **142** is disposed so as to face a ball side of the finger **20**. The adhesive face **144** located on a back face side of the third supporting portion **143** is adhered to the tip of the finger. The adhesive face **144** located on a back face side of the second supporting portion **142** is adhered to the ball side of the finger **20**.

[0064] As illustrated in FIG. **1**, the supporting member **14** includes a second retaining portion **146**. The second retaining portion **146** extends leftward and rightward of the second supporting portion **142**. As illustrated in FIG. **2**, the adhesive face **144** includes a second adhesive face **144d**. The second adhesive face **144d** is located on a back face side of the second retaining portion **146**.

[0065] As illustrated in FIGS. **4** and **5**, the second retaining portion **146** is wrapped around a portion of the supporting member **14** including the first retaining portion **145**. The second adhesive face **144d** is adhered to the surface of the first retaining portion **145** while covering the surface. As a result, the retention of the sensor **10** to the finger **20** is finished.

[0066] In this state, the light emitter **11** and the light detector **12** are disposed so as to face each other with the finger **20** in between. The light emitted from the light emitter **11** that has passed through the first translucent portion **144a** of the supporting member **14** is incident on the tissue of the finger **20**. The light that has passed through the tissue and the second translucent portion **144b** of the supporting member **14** is incident on the light detector face of the light detector **12**. The pulse oximeter performs processing for calculating a light absorber concentration in arterial blood contained in the tissue based on the detection signal outputted from the light detector **12**.

[0067] FIG. **6** illustrates a cross-sectional configuration of the supporting member **14** corresponding to a region VI illustrated in FIG. **1**. The supporting member **14** includes a first substrate **147a** and a second substrate **147b**. The first substrate **147a** is located on the front face side of the supporting member **14**. The second substrate **147b** is located on the back face side of the supporting member **14** so as to form the adhesive face **144**. The light detector **12** is supported between the first substrate **147a** and the second substrate **147b**. Although not illustrated, the light emitter **11** is similarly supported between the first substrate **147a** and the second substrate **147b**.

[0068] The supporting member **14** includes a first surface-treated layer **148**. The first surface-treated layer **148** has at least one different physical property from the first substrate **147a**. In the present exemplary embodiment, the first substrate **147a** and the first surface-treated layer **148** have different surface smoothness. Namely, the surface smoothness is an example of the physical property.

[0069] As used herein, the term “surface treatment” is meant to include any treatment that is capable of adding another layer having a different physical property to the first substrate **147a**. Examples of such a treatment include printing, coating, and adhesion of a film member.

[0070] In the present exemplary embodiment, the first surface-treated layer **148** is not formed in the third supporting portion **143**. In other words, in the third supporting portion **143**, the surface of the first substrate **147a** is exposed. As a result, the rigidity of the third supporting portion **143** is made less than that of an adjacent portion where the first surface-treated layer **148** is formed. In other words, the third supporting portion **143** has higher flexibility than the adjacent portion.

[0071] As illustrated in FIG. **3**, the third supporting portion **143** is a portion that is to be subjected to bending at the tip of the finger **20** of the subject. According to the above configuration, not only the supporting member **14** can be bent with a smaller force, but also the maintainability of the bent state is enhanced. As a result, it is possible to suppress misalignment of the light emitter **11** and the light detector **12** after the sensor **10** is fixed to the finger **20**. Therefore, it is possible to improve the convenience of the sensor **10** configured such that the light emitter **11** and the light detector **12** are attached to the finger **20** with the supporting member **14**.

[0072] Although it is general to form the first surface-treated layer **148** on the surface of the first substrate **147** uniformly in order to provide a desired physical property, in the present exemplary embodiment, local omission of the first surface-treated layer **148** is intentionally performed in order to realize the enhancement of the local flexibility of the third supporting portion **143**. Accordingly, the above-described effect can be obtained while reducing the cost associated with the formation of the first surface-treated layer **148**.

[0073] However, it is not necessarily required to expose the surface of the first substrate **147a** in the third supporting portion **143**. As illustrated in FIG. **7**, the first surface-treated layer **148** may be formed so as to cover the third supporting portion **143** as well. In this case, by providing another first surface-treated layer **148** in a portion adjacent to the third supporting portion **143**, the flexibility of the third supporting portion **143** can be relatively enhanced.

[0074] As illustrated in FIG. **1**, the first retaining portion **145** includes low adhesiveness portion **145a**. The low adhesiveness portion **145a** has lower adhesiveness than an adjacent portion.

[0075] As described with reference to FIGS. **4** and **5**, when the sensor **10** is fixed to the finger, the second adhesive face **144d** located on the back face side of the second retaining portion **146** is adhered to the surface of the first retaining portion **145**. Since a portion having low adhesiveness is formed on the surface, the peelability of the second retaining portion **146** from the first retaining portion **145** can be enhanced. As a result, it is possible to facilitate the operation of re-attaching the sensor **10** in order to, for example, optimize the arrangement of the light emitter **11** and the light detector **12**. Therefore, it is possible to improve the convenience of the sensor **10** configured such that the light emitter **11** and the light detector **12** are attached to the finger **20** with the supporting member **14**.

[0076] In the present exemplary embodiment, similarly to the case of the third supporting portion **143** described with reference to FIG. **6**, the formation of the first surface-treated layer **148** is locally omitted, whereby the low adhesiveness portion **145a** is formed. In other words, the adhesiveness is relatively reduced by exposing the surface of the first substrate **147a** having lower smoothness.

[0077] Although it is general to form the first surface-treated layer **148** on the surface of the first substrate **147** uniformly in order to provide a desired physical property, in the present exemplary embodiment, local omission of the first surface-treated layer **148** is intentionally performed in order to realize the low adhesiveness portion **145a**. Accordingly, the above-described effect can be obtained while reducing the cost associated with the formation of the first surface-treated layer **148**.

[0078] However, it is not necessarily required to expose the surface of the first substrate **147a** in order to form the low adhesiveness portion **145a**. Roughening may be performed on a portion of a first surface-treated layer **148** that is uniformly formed in order to cause the portion to have relatively low adhesiveness. Examples of roughening include printing, coating, adhesion of film members, and the like.

[0079] As an example, the first substrate **147a** may be formed of a nonwoven fabric. In this case, since the first surface-treated layer **148** is formed in order to enhance the smoothness of the surface,

not only the adhesiveness of the portion other than the low adhesiveness portion **145a** can be enhanced, but also the fluffing on the surface of the nonwoven fabric can be suppressed. In order to suppress degradation of the smoothness of the surface, a woven fabric, a film material, a foam material, or the like may be used as the first substrate **147a**.

[0080] As illustrated in FIG. **1**, information for assisting attachment of the sensor **10** to the finger **20** is provided on the front face side of the supporting member **14**. Such information includes information indicating the position of the light emitter **11**, information indicating the position of the light detector **12**, information providing a guide indicating the position at which the finger **20** is to be placed, and the like.

[0081] As illustrated in FIG. **6**, the information is formed by a second surface-treated layer **149**. The second surface-treated layer **149** may be formed on at least one of the surface of the first substrate **147a** and the surface of the first surface-treated layer **148**.

[0082] According to the above configuration, it is possible to improve the convenience of attachment of the sensor **10** to the finger **20**, while ensuring the above-described advantage related to the high flexibility and the low adhesiveness.

[0083] In the present exemplary embodiment, the surface of the first surface-treated layer **148** is formed so as to exhibit a different appearance from of the surface of the first substrate **147a**. As a result, the third supporting portion **143** and the low adhesiveness portion **145a** on which the first surface-treated layer **148** is not formed have different appearance from other portions of the supporting member **14**.

[0084] According to such a configuration, since it is possible to emphasize, to the user, a portion capable of providing the above-described advantage, it is possible to enhance the merchantability of the sensor **10**.

[0085] The difference in appearance between the first substrate **147a** and the first surface-treated layer **148** need not be provided by the difference in materials forming the respective elements. The difference in appearance may be provided by a color, a pattern, or the like.

[0086] FIG. **8** illustrates another exemplary configuration of the sensor **10**. FIG. **9** illustrates a sectional configuration of the supporting member **14** corresponding to a region IX illustrated in FIG. **8**. In this example, the flexibility of each of the first supporting portion **141** and the second supporting portion **142** of the supporting member **14** is made higher than an adjacent portion.

[0087] Specifically, the first surface-treated layer **148** is not formed in the first supporting portion **141** and the second supporting portion **142** (the first supporting portion **141** is not illustrated). As a result, the rigidity of each of the first supporting portion **141** and the second supporting portion **142** is made less than that of the adjacent portion where the first surface-treated layer **148** is formed.

[0088] The first supporting portion **141** and the second supporting portion **142** are portions that support and cover the light emitter **11** and the light detector **12**, respectively. According to such a configuration, elongation of the first substrate **147a** due to the coverage of the light emitter **11** and the light detector **12** is promoted, so that it is possible to suppress generation of a stress as well as the misalignment of the light emitter **11** and the light detector **12** due to such stress. Therefore, it is possible to improve the convenience of the sensor **10** configured such that the light emitter **11** and the light detector **12** are attached to the finger **20** with the supporting member **14**.

[0089] Although it is general to form the first surface-treated layer **148** on the surface of the first substrate **147** uniformly in order to provide a desired physical property, in the present exemplary embodiment, local omission of the first surface-treated layer **148** is intentionally performed in order to realize the enhancement of the local flexibility of the first supporting portion **141** and the second supporting portion **142**. Accordingly, the above-described effect can be obtained while reducing the cost associated with the formation of the first surface-treated layer **148**.

[0090] However, it is not necessarily required to expose the surface of the first substrate **147a** in the first supporting portion **141** and the second supporting portion **142**. As illustrated in FIG. **10**, a first surface-treated layer **148** may be formed so as to cover the second supporting portion **142** as

well (the same applies to the first supporting portion **141**). In this case, by providing another first surface-treated layer **148** in a portion adjacent to each of the first supporting portion **141** and the second supporting portion **142**, the flexibility of these portions can be relatively enhanced. [0091] Each of the configurations exemplified above is merely illustrative for facilitating understanding of the present disclosure. The configuration examples may be appropriately modified or combined with other configuration examples without departing from the spirit of the present disclosure.

[0092] The portion in which the flexibility is made higher than the adjacent portion in the supporting member **14** may be at least one of the first supporting portion **141**, the second supporting portion **142**, and the third supporting portion **143**, depending on the specification of the sensor **10**.

[0093] As illustrated in FIG. **1**, the supporting member **14** according to the above exemplary embodiment has a shape that is axisymmetric with respect to a straight line L connecting the light emitter **11** and the light detector **12**. According to such a configuration, the attachment stability of the sensor with respect to the finger of the subject can be enhanced.

[0094] However, as illustrated in FIG. **11**, the supporting member **14** may exhibit a belt shape in which the light emitter **11** is supported at one end thereof. In this example, fixation of the sensor **10** is performed by wrapping the supporting member **14** around the finger of the subject. The light detector **12** is supported at a position facing the light emitter **11** with the finger in between when the supporting member **14** is wrapped around the finger.

[0095] The application of the medical sensor to which the concept of the present disclosure may be applied is not limited to the detection of the blood light absorber concentration. In the case of a medical sensor that optically detects a physiological parameter, the fundamental concept of the present disclosure can be applied through an appropriate change of the shape.

Claims

1. A medical sensor, comprising: at least one light emitter configured to emit light; at least one light detector configured to output a signal corresponding to an amount of incident light; and a supporting member adapted to be attached to a body of a subject, wherein the supporting member includes: a first supporting portion supporting and covering the light emitter; a second supporting portion supporting and covering the light detector; and a third supporting portion located between the first supporting portion and the second supporting portion; and wherein at least one of the first supporting portion, the second supporting portion and the third supporting portion includes a portion with higher flexibility than an adjacent portion thereof.
2. The medical sensor according to claim 1, wherein the supporting member includes: a substrate; and a first surface-treated layer formed on a surface of the substrate and having a different physical property from the substrate, wherein the first surface-treated layer is not formed at the portion with higher flexibility.
3. The medical sensor according to claim 2, wherein the first surface-treated layer is so formed as to exhibit a different appearance from the substrate.
4. The medical sensor according to claim 2, wherein the supporting member includes a second surface-treated layer formed on a surface of at least one of the substrate and the first surface-treated layer so as to provide information for assisting attachment to the body of the subject.
5. The medical sensor according to claim 2, wherein the substrate is formed with nonwoven fabric.
6. The medical sensor according to claim 1, wherein the supporting member exhibits an axisymmetric shape relative to a straight line connecting the light emitter and the light detector.
7. A medical sensor, comprising: at least one light emitter configured to emit light; at least one light detector configured to output a signal corresponding to an amount of incident light; and a supporting member adapted to be attached to a body of a subject, wherein the supporting member

includes: a first retaining portion having a first adhesive face adapted to be adhered to the body; and a second retaining portion having a second adhesive face adapted to be adhered to a surface of the first retaining portion while covering the surface, wherein at least a part of the surface includes a portion with lower adhesiveness than an adjacent portion thereof.

8. The medical sensor according to claim 7, wherein the supporting member includes: a substrate; and a first surface-treated layer formed on a surface of the substrate and having a different physical property from the substrate, wherein the first surface-treated layer is not formed at the portion with lower adhesiveness.

9. The medical sensor according to claim 8, wherein the first surface-treated layer is so formed as to exhibit a different appearance from the substrate.

10. The medical sensor according to claim 8, wherein the supporting member includes a second surface-treated layer formed on a surface of at least one of the substrate and the first surface-treated layer so as to provide information for assisting attachment to the body of the subject.

11. The medical sensor according to claim 8, wherein the substrate is formed with nonwoven fabric.

12. The medical sensor according to claim 7, wherein the supporting member exhibits an axisymmetric shape relative to a straight line connecting the light emitter and the light detector.

13. A sensor supporting member adapted to attach, to a body of a subject, a medical sensor provided with at least one light emitter configured to emit light and at least one light detector configured to output a signal corresponding to an amount of incident light, the sensor supporting member comprising: a first supporting portion supporting and covering the light emitter; a second supporting portion supporting and covering the light detector; and a third supporting portion located between the first supporting portion and the second supporting portion, wherein at least one of the first supporting portion, the second supporting portion and the third supporting portion includes a portion with higher flexibility than an adjacent portion thereof.

14. A sensor supporting member adapted to attach, to a body of a subject, a medical sensor provided with at least one light emitter configured to emit light and at least one light detector configured to output a signal corresponding to an amount of incident light, the sensor supporting member comprising: a first retaining portion having a first adhesive face adapted to be adhered to the body; and a second retaining portion having a second adhesive face adapted to be adhered to a surface of the first retaining portion while covering the surface, wherein at least a part of the surface includes a portion with lower adhesiveness than an adjacent portion thereof.
